

MOSAM BISWAS

How *mosambiswas.com* is built — the typography, the palette, the hand-rolled code, and the *bubbles*. A design document for a site with *zero frameworks* and one vermilion accent.

STACK

HTML5 · CSS3 · Vanilla JS

DESIGN

Editorial dark · hairline grid

SIGNATURE

The bubble cursor ●

ZERO *frameworks*

The whole site is three hand-written files — *index.html*, *style.css*, *main.js* — served as static pages. No build step, no bundler, no icon packs. Every interaction is written from scratch.

The only external dependency is Google Fonts. Everything else — the cursor, the reveals, the preloader, the parallax — is vanilla JavaScript inside a single IIFE with 'use strict', gated behind capability checks (**pointer: fine**, **prefers-reduced-motion**) so the site degrades gracefully on touch devices, slow connections, and with JavaScript disabled.

FRONTEND	Pure HTML5, CSS3 (~1,450 lines), Vanilla JS (~450 lines) — no libraries
HOSTING	GitHub Pages + custom domain via CNAME (www.mosambiswas.com)
PWA	manifest.json + service worker: offline cache, SKIP_WAITING auto-update, reload on controllerchange
SEO	JSON-LD (WebSite + Person), Open Graph, Twitter cards, sitemap.xml, robots.txt, canonical URL
A11Y	Semantic HTML, skip-link, sr-only h1, focus-visible rings, reduced-motion support, no-JS fallback
LICENSE	MIT — open as a template for other portfolios

FILE ANATOMY

```
BiswasMosam.github.io/
├─ index.html      # one page, five sections, ASCII-art easter egg
├─ style.css       # the whole design system
├─ main.js         # preloader · cursor · reveals · parallax · clock
├─ service-worker.js # offline + silent updates
├─ certificates.html # certificate gallery + modal
└─ sheichobi/      # photography portfolio ("that picture" in Bengali)
```

09

PROJECTS LISTED

01

IEEE PUBLICATION

100

LIGHTHOUSE PERF

0kb

JS DEPENDENCIES

TYPE IS THE *interface*

Four Google Fonts, each with a strict role. Display type is fitted **edge-to-edge by JavaScript**: *fitLines()* measures each hero word at 100px, then scales the font-size by the ratio of line width to word width — re-run on font load and debounced resize, so “MOSAM” and “BISWAS” always touch both margins.

SYNE 700-800

DISPLAY — HEADINGS, HERO, MARQUEE

Uppercase, tracking -0.02em, line-height 0.92-1.05.
The loud voice.

Instrument Serif

ACCENT — EM WORDS, SECRET LAYERS

Italic only, often in vermillion. Marks everything human or hidden.

Manrope 400-600

BODY — PARAGRAPHS, FACTS, LEDES

1rem / 1.6 line-height, dimmed to 58% ink for long text.

SPACE MONO

LABELS — .MONO UTILITY CLASS

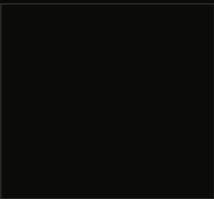
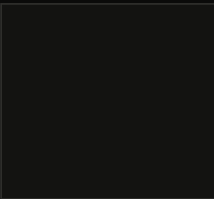
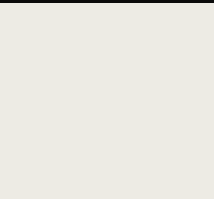
0.72rem, uppercase, +0.14em tracking. Indexes, clocks, whispers.

EDITORIAL DEVICES

Section headers follow a fixed grammar — a numbered mono label “(01) — **Selected Work**” above a Syne title where one word slips into *serif italic*. Nav links carry superscript counts (Work⁰⁹), the second hero line renders as **outline text** via `-webkit-text-stroke` with a solid-color @supports fallback, and hairlines — 1px at 14% ink — rule every section like a printed spread.

WARM BLACK, *one accent*

The palette is deliberately narrow: a warm near-black, a warm off-white, and a single vermilion. Everything in between is the same ink at reduced alpha — so the page reads like one material lit at different intensities, never a rainbow of grays.

					
#0b0b0a	#131311	#edebe4	ink · 58%	ink · 34%	#ff5227
--BG	--BG-2	--INK	--INK-DIM	--INK-FAINT	--ACCENT

THE SYSTEM IN SIX CUSTOM PROPERTIES

```
:root {
  --bg: #0b0b0a; --bg-2: #131311; --ink: #edebe4;
  --line: rgba(237, 235, 228, 0.14); /* hairline grid */
  --accent: #ff5227; /* the only color */
  --pad: clamp(1.25rem, 4vw, 4rem); /* fluid gutters */
  --ease: cubic-bezier(0.22, 1, 0.36, 1); /* one shared easing */
}
```

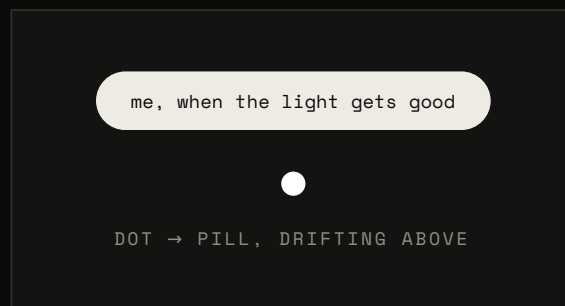
HAIRLINES	1px borders at 14% ink separate every section, row and fact — the “printed grid” feel
FLUID SCALE	Every size is a clamp() — hero name clamp(3.4rem, 15.5vw, 15rem), section titles clamp(2.4rem, 6vw, 4.6rem)
ONE EASING	A single expo-out curve (0.22, 1, 0.36, 1) drives reveals, hovers, menu, preloader — motion feels like one hand
FILM GRAIN	Fixed full-screen SVG feTurbulence noise (data-URI) at 4.5% opacity — analog texture, zero image requests
BLEND MODES	Header and cursor use mix-blend-mode: difference, staying legible over any background
SELECTION	::selection inverts to vermilion — even highlighting text is on-brand; scrollbars hidden site-wide

A CURSOR THAT *talks*

One 12px dot follows the pointer through a lerp loop ($x += (target - x) \times 0.22$, per animation frame). It never stays a dot: it morphs into *whisper pills* and a *reading lens* depending on what — even which *word* — you hover.

MODE 1 — WHISPER

Any element with a **data-whisper** attribute makes the dot stretch into an ink-colored caption pill with a Space Mono aside. It reacts to **separate words**: in “Developer, researcher, photographer” each word carries its own whisper (“*me, when a keyboard is in reach*”, “*me, when a question won’t let go*”, “*me, when the light gets good*”). A **||** in the attribute separates variants that cycle on every re-hover. The pill drifts 26px above the pointer, flips below near the top edge, and clamps itself inside the viewport near the sides. Width is measured from the label + 36px.



MODE 2 — LENS

Paragraphs marked **data-secret** hold two layers: the visible text and a hidden **.secret__layer** — the same thought rewritten candidly, in accent-orange Instrument Serif italic. Over these, the bubble becomes a 78px-radius transparent circle with a vermilion border, and the hidden layer is revealed through a per-frame **clip-path: circle(r at x y)** that tracks the cursor — a reading glass that shows what the page really thinks. Radius eases at 0.16/frame, so the lens breathes open and closed.



UNDER THE HOOD

```
/* one rAF loop drives everything */
x += (targetX - x) * 0.22;           // dot follows with inertia
pillOffsetY += (offsetTarget - pillOffsetY) * 0.2; // whisper drift (26px up, -34 near top)
lensR += (lensTargetR - lensR) * 0.16; // lens radius easing → 78px
layer.style.clipPath = `circle(${lensR}px at ${x-rect.left}px ${y-rect.top}px)`;
```

Links grow the dot to a 52px halo in **mix-blend-mode: difference**; links with their own whisper show the pill instead. The whole system mounts only on fine pointers and switches off under prefers-reduced-motion — mobile falls back to the native cursor, nothing breaks.

ALIVE IN THE *details*

PRELOADER	A counter eases 00 → 100 with cubic ease-out over 1s (rAF), then the curtain slides up with translateY(-100%) and the masked hero type reveals
TYPE REVEAL	Hero words sit at translateY(112%) inside overflow-hidden lines; loading flips them to 0 with a 0.12s stagger — the classic masked reveal, no library
SCROLL REVEALS	IntersectionObserver (threshold 0.12, -6% bottom margin) adds .is-in once per element; gated behind an html.js class so no-JS users see everything instantly
WORK ROWS	Hover inverts each row via a ::before that scaleY's from bottom, flipping text to background color while the index digit turns vermilion and the title slides 10px
PROJECT PREVIEW	A cursor-following card (lerp 0.12) shows a vertical strip of 9 gradient figures; hovering row n translates the strip by n × -100% — desktop only
PARALLAX	Images with data-parallax get translateY(offset × -speed) + scale(1.12) from a scroll-throttled rAF, skipping anything off-screen
MARQUEE	Pure CSS: a duplicated track animating translateX(-50%) over 36s — infinite skills ticker with zero JS
LIVE CLOCK	Intl.DateTimeFormat pinned to Asia/Kolkata ticks every second in header and footer — the site knows what time it is at home
COPY EMAIL	navigator.clipboard with an execCommand('copy') textarea fallback; the button flips to "Copied ✓" for 1.4s
MENU	Fullscreen overlay slides down with staggered per-link delays (0.10s → 0.34s); Escape closes, body scroll locks
CERT MODAL	Certificates open in a dimmed dialog (rgba(8,8,7,0.96)); close by ×, backdrop click, or Escape — the × rotates 90° on hover
PORTRAIT	About photo is sticky and grayscale(1); hovering restores color — a small reward for reading

EASTER EGGS

A console signature greets anyone who opens DevTools (*"thanks for peeking under the hood"*), a giant ASCII-art portrait hides in an HTML comment at the end of index.html, the résumé whispers *"aka Janam Kundali"*, and the footer credit insists — correctly — that *"even this bubble is handmade."*

FAST BY *subtraction*

ZERO DEPS	No framework, no icon font, no analytics script — the only third-party bytes are four Google Fonts with preconnect hints
GPU MOTION	Every animation is transform/opacity only — reveals, marquee, parallax, cursor — so nothing forces layout or paint
RAF DISCIPLINE	Scroll and mousemove handlers are passive and coalesced into requestAnimationFrame; resize work is debounced
IMAGES	Photography ships as WebP with lazy loading + async decoding; previews are CSS gradients instead of images
SERVICE WORKER	Caches the shell for offline visits; new deploys post SKIP_WAITING and the page reloads once on controllerchange — updates are silent
REDUCED MOTION	prefers-reduced-motion collapses all animation to 0.01ms, kills the cursor and marquee, and shows content immediately
NO-JS	Reveal styles only apply under an html.js flag set by script — without JavaScript the site is simply... all visible
RESULT	100 Lighthouse performance, sub-second loads, installable as a PWA

(-) COLOPHON

LET'S *talk*

This document mirrors the site it describes — same warm black, same four typefaces, same hairlines, same single accent. Built from the live source of **Portfolio v5**, July 2026.

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